

dTDP-L-Rhamnose Biosynthesis: A Concise Review of a Deeply Conserved Bacterial Nucleotide-Sugar Module

A commissioned review-style synthesis for a molecular-biology audience. Search date: 2026-07-16. Evidence base: 29 primary papers and reviews (PubMed).

1. Executive Summary

dTDP-L-rhamnose biosynthesis is one of the best-characterized nucleotide-sugar supply modules in bacteria: a compact, cytoplasmic, four-enzyme cassette (RmlA, RmlB, RmlC, RmlD — historically also named RfbA/RffH, RffG/RfbB, RmlC/RfbC, and RmlD/RfbD) that converts α -D-glucose-1-phosphate and dTTP into the activated glycosyl donor dTDP- β -L-rhamnose. The pathway runs as a strictly ordered linear sequence: **thymidylyl transfer (RmlA) $\rightarrow$ 4,6-dehydration (RmlB) $\rightarrow$ 3,5-double epimerization (RmlC) $\rightarrow$ 4-ketoreduction (RmlD)**. Each step was defined biochemically by reconstitution in *Salmonella enterica* serovar Typhimurium LT2, and the enzymes' three-dimensional structures and mechanisms are now known in detail, making this pathway an unusually complete case study in nucleotide-sugar chemistry.

The order of reactions is dictated by chemistry rather than by convention. RmlB installs a 4-keto group and a 6-deoxy function by a self-contained oxidation–dehydration–reduction cascade driven by a tightly bound, recycled NAD⁺. That 4-keto group is the mechanistic handle RmlC needs to abstract and re-deliver protons at C3' and C5', and it is removed only at the very last step, when RmlD reduces the 4-carbonyl using NAD(P)H. Because the epimerization is thermodynamically unfavorable on its own, the irreversible RmlD reduction pulls the whole pathway forward. RmlA sits at the top of the module as its committed, regulated entry point: it is a homotetramer that is feedback-inhibited by the end product dTDP-L-rhamnose, giving the module intrinsic self-regulation, and it can be shut down by nanomolar allosteric inhibitors that act at a remote second site.

The module's boundaries are sharp. It ends with the release of soluble dTDP- β -L-rhamnose; all downstream O-antigen, LPS core, capsule, rhamnan, and rhamnolipid assembly reactions are consumers, not part of the biosynthetic module. The cassette is essential wherever rhamnose is

a required surface-glycan component, and — critically — it has no human counterpart: eukaryotes reach an activated rhamnose (UDP-L-rhamnose, not dTDP-L-rhamnose) by a convergent single-polypeptide route. This absence in humans, combined with essentiality and virulence relevance in pathogens such as *Streptococcus pyogenes*, *Streptococcus mutans*, *Mycobacterium tuberculosis*, and *Pseudomonas aeruginosa*, makes RmlA–RmlD a validated, selective antibacterial target set.

2. Definition and Biological Boundaries

What is included

The dTDP-L-rhamnose module comprises exactly four enzymatic activities operating on soluble, cytoplasmic nucleotide-sugar intermediates:

Step	Enzyme (synonyms)	Activity	Substrate → Product
1	RmlA (RfbA/RffH) — glucose-1-phosphate thymidyltransferase	Nucleotidyltransferase	α -D-glucose-1-P + dTTP → dTDP-D-glucose + PP _i
2	RmlB (RffG/RfbB) — dTDP-glucose 4,6-dehydratase	Oxidoreductase/lyase (internal NAD ⁺)	dTDP-D-glucose → dTDP-4-keto-6-deoxy-D-glucose + H ₂ O
3	RmlC (RfbC) — dTDP-4-dehydrorhamnose 3,5-epimerase	Cofactor-independent bis-epimerase	dTDP-4-keto-6-deoxy-D-glucose → dTDP-4-keto-L-rhamnose
4	RmlD (RfbD) — dTDP-4-dehydrorhamnose reductase	NAD(P)H-dependent SDR reductase	dTDP-4-keto-L-rhamnose → dTDP-β-L-rhamnose

The scope is the *supply* of the activated sugar. The module's product, dTDP- β -L-rhamnose, is the shared glycosyl donor pool from which rhamnosyltransferases draw.

What should be treated separately

Several neighboring processes are frequently conflated with the module but lie outside it:

- **Downstream glycan assembly.** O-antigen polymerization, LPS core decoration, capsule/Group A carbohydrate synthesis, rhamnan wall polysaccharide, and rhamnolipid surfactant

assembly all consume dTDP-L-rhamnose but are catalyzed by distinct rhamnosyltransferases and polymerases. In *P. aeruginosa*, L-rhamnose supplied by the *rml* locus is the LPS-core attachment point for O antigen (PMID: 11065359), but that attachment chemistry is not part of the biosynthetic module.

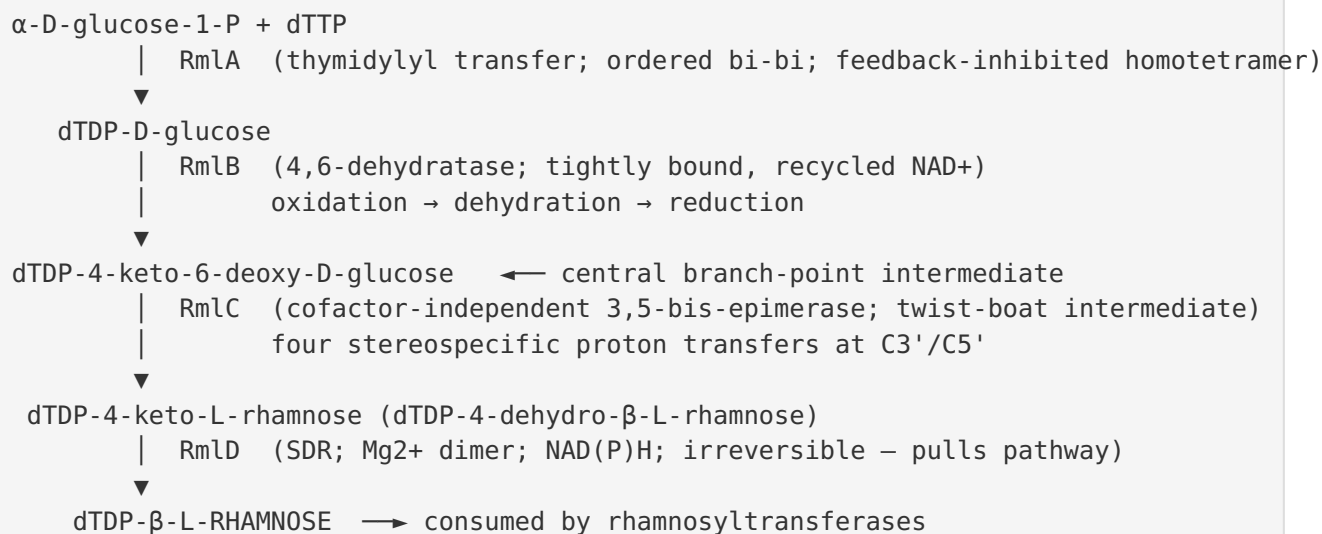
- **Other NDP-deoxysugar branches.** The RmlA and RmlB reactions are shared priming steps that feed many specialized deoxysugar pathways (e.g., dTDP-D-Qui4NFo; PMID: 23418147). The point at which a pathway "belongs" to rhamnose versus another product is the RmlC/RmlD step.
- **The eukaryotic UDP-rhamnose route.** Plants and other eukaryotes make an activated rhamnose, but on a UDP rather than dTDP carrier and via a single bifunctional enzyme (see §5). This is a convergent solution and should not be described as the same system.

Competing definitions and nomenclature confusion

A recurring source of literature confusion is **synonymous gene naming**. The same four activities appear under the *rml* nomenclature and under the older *rfb/rff* nomenclature inherited from O-antigen (*rfb*) and enterobacterial common antigen (*rff*) gene clusters. In *S. mutans*, the genes were originally identified and named by homology to *rfbA*, *rfbB*, and *rfbD* products involved in dTDP-L-rhamnose anabolism from D-glucose-1-phosphate (PMID: 9023194). Readers must map, for example, *rffG* → RmlB and *rfbC* → RmlC to avoid double-counting or mis-ordering enzymes. The review brief's provisional outline (*rfbA* / *rffG* / *rmlC* / *rfbC* / *rfbD* / PP_0500) reflects exactly this heterogeneous naming, and all of those labels resolve onto the same four-activity cassette.

3. Mechanistic Overview

The best-supported model is a **strictly ordered, linear, four-step pathway** established by overexpression and reconstitution of all four enzymes from *S. enterica* sv. Typhimurium LT2, which also assigned the terminal two steps: RmlC as the NAD(P)H-independent 3,5-epimerase and RmlD as the NAD(P)H-dependent 4-reductase (PMID: 10455186).



Obligatory ordering. The sequence cannot be permuted. RmlB must act before RmlC because the 4-keto group that RmlB generates is the electron sink that makes the C3'/C5' protons acidic enough for RmlC to abstract. RmlD must act last because it destroys that same 4-keto group by reduction; if it acted earlier there would be no keto handle for epimerization. Thus RmlB → RmlC → RmlD is a chemically enforced order, not merely an observed one.

Thermodynamic pull. RmlC epimerization is unfavorable in isolation — equilibrium lies far toward the D-configured substrate (~3% conversion). The pathway is driven forward by the effectively irreversible RmlD reduction, which continually removes the L-configured 4-keto product. This is a classic example of an unfavorable stereochemical step coupled to an irreversible downstream reaction.

Regulation. The committed step is RmlA. Feedback inhibition of RmlA by dTDP-L-rhamnose provides intrinsic homeostatic control of L-rhamnose output ([PMID: 11118200](#)). Accessory (non-obligatory) features include the allosteric second site on RmlA exploited by synthetic inhibitors ([PMID: 23138692](#)).

4. Major Molecular Players and Active Assemblies

RmlA — glucose-1-phosphate thymidyltransferase (the regulated gate)

RmlA catalyzes thymidyl transfer from dTTP to glucose-1-phosphate by an **ordered bi-bi mechanism**, releasing pyrophosphate and dTDP-glucose. The first crystal structures of this enzyme class (Ep/RmlA) with dTTP and product clarified the nucleotidyltransferase catalytic

mechanism and enabled protein engineering to accept "unnatural" sugar phosphates, a foundation for glycorandomization (PMID: 11373625). The *P. aeruginosa* enzyme was solved to 1.66 Å and is a **homotetramer** whose monomer has three subdomains: a core nucleotidyltransferase subdomain shared with other NTP-transferases, plus RmlA-specific sugar-binding and dimerization subdomains. Five substrate/product complexes defined catalysis, and a dTDP-L-rhamnose complex revealed the structural basis of **end-product feedback inhibition** — "RmlA is inhibited by dTDP-L-rhamnose thereby regulating L-rhamnose production in bacteria" (PMID: 11118200). Its druggability is notable: nanomolar inhibitors bind a **second site remote from the active site** yet act as competitive inhibitors of glucose-1-phosphate with high cooperativity, working by blocking the conformational change central to the ordered bi-bi mechanism (PMID: 23138692).

RmlB — dTDP-glucose 4,6-dehydratase (internal-cofactor cascade)

RmlB carries out a three-part reaction — oxidation, dehydration, and reduction — using a **tightly bound NAD⁺ as a recycled internal cofactor** rather than consuming external redox equivalents. The 1.5 Å structure of *Streptococcus suis* RmlB, with the cofactor trapped as NADH, shows the sequence: "NAD(+) abstracts a hydride from the C4' atom of dTDP-glucose-forming NADH. After elimination of water, hydride is then transferred back to the C6' atom of dTDP-4-keto-5,6-glucosene-regenerating NAD(+)" (PMID: 14505409). The reduced nicotinamide adopts a distorted boat conformation that tunes its hydride-donor ability. Because NAD⁺ is regenerated each turnover, RmlB is formally a lyase despite its redox chemistry.

RmlC — dTDP-4-dehydrorhamnose 3,5-epimerase (cofactor-free double inversion)

RmlC is the smallest enzyme of the cassette (~22 kDa) and is remarkable for using **no cofactor and no metal** — its activity was established to be NAD(P)H-independent in the original reconstitution (PMID: 10455186). It catalyzes **four stereospecific proton transfers** to invert configuration at both C3' and C5', converting the D-configured 4-keto sugar into the L-configured 4-keto-rhamnose; structural and deuterium-exchange work shows that "RmlC catalyzes four stereospecific proton transfers and the substrate undergoes a major conformational change during the course of the transformation," passing through an unusual **twist-boat conformation** (PMID: 17046787). RmlC adopts a **jelly-roll (cupin) fold**; the archaeal ortholog demonstrates deep conservation of both fold and active site (see §5).

RmlD — dTDP-4-dehydrorhamnose reductase (atypical SDR, Mg²⁺-dependent dimer)

RmlD completes the pathway by NAD(P)H-dependent reduction of the 4-carbonyl to give dTDP-β-L-rhamnose. Structures of *Salmonella* RmlD with NADH, NADPH, and product reveal a **short-chain dehydrogenase/reductase (SDR) fold** with a **novel Mg²⁺-containing dimer interface** — a feature not seen in canonical SDRs: "RmlD differs from other short chain dehydrogenases in that it has a novel dimer interface that contains Mg²⁺. Enzyme catalysis involves hydride transfer from the nicotinamide ring of the cofactor to the C4'-carbonyl group of the substrate. The substrate is activated through protonation by a conserved tyrosine" (PMID: 12057193). This Ser/Thr-Tyr-Lys catalytic triad is conserved across RmlD orthologs (PMID: 10455186). RmlD's step is effectively irreversible and provides the thermodynamic pull for the whole module.

5. Evolutionary and Cell-Biological Variation

A single non-redundant cassette in most bacteria

In organisms where rhamnose is a required cell-wall component, the cassette is typically present as a single copy of each activity. *M. tuberculosis*, for instance, carries genes for "only one isotype each of RmlA to RmlD" (PMID: 11302803), which is part of what makes the pathway an attractive, non-redundant drug target.

Mosaic evolution and horizontal gene transfer

The *rml* genes have a **mosaic evolutionary history** driven by their genomic location within or adjacent to O-antigen gene clusters. In *S. enterica*, "the *rml* gene set of *S. enterica* includes genes with two different evolutionary histories": the 5' region (*rmlB*, *rmlD*, most of *rmlA*) is subspecies-specific and GC-rich, while the 3' region (part of *rmlA* and all of *rmlC*) is O-antigen-specific, lower in GC content, and far more variable — the hallmark of recombination and lateral transfer (PMID: 10974117). The same abrupt conserved-5'/divergent-3' gradient with a GC-content shift recurs in *Vibrio cholerae*, which shows "highly similar sequences at the 5' end *rmlB* gene, but very divergent and strain-specific sequences at the 3' end of the *rml* gene set," their position at the end of O-antigen clusters facilitating lateral transfer (PMID: 12949172). The practical implication is that *rmlC* in particular co-evolves with the serotype-defining O-antigen machinery it feeds.

The convergent eukaryotic route

Eukaryotes solve the same chemical problem — making an activated L-rhamnose — with entirely different molecular means. In *Arabidopsis*, "RHAMNOSE BIOSYNTHESIS1 (RHM1) encodes a UDP-L-rhamnose synthase, and rhm1 mutants exhibit many developmental defects" arising from reduced rhamnose-containing pectic polysaccharides and flavonol glycosides (PMID: 29505161). Biotechnological UDP-rhamnose supply for flavonoid glycosylation likewise relies on plant RHM synthases such as AtRHM2 that convert UDP-glucose to UDP-rhamnose (PMID: 32693583; PMID: 26433966). Two differences matter: the **sugar carrier is UDP, not dTDP**, and the **four bacterial activities are collapsed into one polypeptide**. This is textbook convergent evolution, and it is precisely why the bacterial dTDP route is selectively targetable — humans and plants simply do not run it.

Deep origin of the priming node and radiation of the tailoring steps

The **RmlA (nucleotidyltransferase) + RmlB (4,6-dehydratase)** pair constitutes an ancient, widely shared **deoxysugar-priming node** that feeds numerous NDP-deoxyhexose branches beyond rhamnose (e.g., the dTDP-D-Qui4Nfo pathway begins RmlA → RmlB → VioA → VioF; PMID: 23418147). The **RmlC/RmlD-type tailoring activities have radiated** into many specialized 6-deoxyhexose pathways. An RmlC-type 3,5-epimerase from the archaeon *Methanobacterium thermoautotrophicum* adopts the same jelly-roll (cupin) fold with fully conserved active-site residues: "The active site... is formed by residues that are conserved in all known RmlC sequence homologues. The conservation of the active site residues suggests that the mechanism of action is also conserved" (PMID: 10827167) — implying a deep evolutionary origin for this fold. Homologs of RmlC/RmlD (EvaD, ChmJ, StrM, WbiB) support divergent products including streptose and 6-deoxytalose, and their biosynthesis "has been proposed to follow a similar path to that of TDP-rhamnose, catalyzed by the enzymes RmlA, RmlB, RmlC, and RmlD" (PMID: 35398092; PMID: 21640586). For understanding the ancestral role, the canonical *Salmonella/Pseudomonas/M. tuberculosis* RmlA–D enzymes — being single-copy, non-specialized, and biochemically reconstituted — are the best representatives.

Alternative routes to the same or related products

There is at least one well-documented bacterial alternative that reaches dTDP-L-rhamnose (or its close relative dTDP-6-deoxy-L-talose) by different means: in *Burkholderia thailandensis*, the epimerase **WbiB interconverts dTDP-L-rhamnose and dTDP-6-deoxy-L-talose**, and an RmlA/

RmlB/RmlC/Tal/WbiB combination produces dTDP-L-rhamnose just as RmlA/RmlB/RmlC/RmlD does (PMID: 21640586). This shows that the terminal tailoring chemistry can be achieved by more than one enzymatic logic even within bacteria.

6. Constraints, Dependencies, and Failure Modes

Ordering constraints (chemically enforced)

1. **RmlA first** — no dTDP-sugar exists until thymidylyl transfer occurs; this is also the committed, feedback-regulated step.
2. **RmlB before RmlC** — RmlC epimerization requires the C4' keto group that only RmlB installs (the keto group acidifies the C3'/C5' protons).
3. **RmlD last** — RmlD reduces and thereby *removes* the 4-keto group; performing it earlier would abolish the substrate feature RmlC depends on.

These constraints rule out otherwise-imaginable orderings (e.g., reduce-then-epimerize, or epimerize-before-dehydrate). The evidence against them is mechanistic: the 4-keto group is simultaneously the product of RmlB, the substrate requirement of RmlC, and the target of RmlD.

Thermodynamic dependency

The unfavorable equilibrium of RmlC (~3% conversion in isolation) means the pathway **depends on RmlD's irreversible reduction** to proceed. In vitro, RmlC assays are typically run coupled to RmlD for this reason — a genuine dependency, not merely a convenience.

Cofactor and metal dependencies

- RmlB requires its **tightly bound NAD⁺**, which is recycled and not consumed.
- RmlD requires **NAD(P)H** as a consumed reductant and **Mg²⁺** for its dimer interface.
- RmlC uniquely requires **neither cofactor nor metal**.

Genetic failure modes and essentiality

Loss of any single enzyme abolishes dTDP-L-rhamnose output and, downstream, rhamnose-dependent surface glycans. Insertional inactivation of individual dTDP-rhamnose genes in *S. mutans* eliminates cell-wall rhamnose and the serotype-specific antigen (PMID: 9209063; PMID: 9023194). Across pathogens, "Biosynthesis of the nucleotide sugar precursor dTDP-L-rhamnose

is critical for the viability and virulence of many human pathogenic bacteria, including *Streptococcus pyogenes* (Group A Streptococcus; GAS), *Streptococcus mutans* and *Mycobacterium tuberculosis*" (PMID: 30600561). In *P. aeruginosa*, *rmlC* mutants make a truncated LPS core that cannot anchor O antigen (PMID: 11065359) — a concrete surface-assembly failure mode traceable directly to the module.

Compartmental constraint

The entire module is **cytoplasmic and soluble**; its product is delivered to membrane-associated glycosyltransferases. There is no evidence that the four enzymes assemble into an obligate physical channel — the intermediates are diffusible dTDP-sugars — although co-localization within *rml* operons is common and may favor efficient hand-off.

7. Controversies and Open Questions

Strongly supported claims. The four-step order, the assignment of each activity, and the individual enzyme mechanisms are all backed by direct structural and biochemical evidence: RmlA's tetrameric structure and feedback inhibition (PMID: 11118200); RmlB's recycled-NAD⁺ cascade caught crystallographically as NADH (PMID: 14505409); RmlC's cofactor-free twist-boat double epimerization (PMID: 17046787); and RmlD's Mg²⁺-dimer SDR mechanism (PMID: 12057193). The pathway's essentiality and its absence from humans are also well established.

Areas of weaker or indirect evidence, and open questions:

1. **Cross-organism mixing.** Much of the mechanistic detail is drawn from different organisms — RmlA from *P. aeruginosa*, RmlB from *S. suis*, RmlC/RmlD from *Salmonella* and archaea. While fold and active-site conservation argue that these are legitimately comparable, quantitative kinetics and regulation may differ by species, and reviews sometimes over-generalize from one enzyme to "the pathway."
2. **NADH vs NADPH preference of RmlD.** RmlD structures were obtained with both NADH and NADPH (PMID: 12057193); the physiological cofactor preference (and whether it varies across bacteria) is not fully settled.
3. **Extent of feedback regulation.** dTDP-L-rhamnose feedback inhibition of RmlA is demonstrated structurally in *P. aeruginosa* (PMID: 11118200), but whether this is the dominant in vivo control point across all rhamnose-producing bacteria — versus transcriptional control of the *rml* operon — is less clear.

4. **Boundary with tailoring pathways.** Because RmlA/RmlB are shared and RmlC/RmlD homologs radiate into many products, defining exactly where "rhamnose biosynthesis" ends and a "specialized deoxysugar pathway" begins is partly conventional; enzymes like WbiB blur this line ([PMID: 21640586](#)).
5. **Druggability translation.** Potent in vitro inhibitors of RmlA and the other enzymes exist ([PMID: 23138692](#); [PMID: 30600561](#)), but whole-cell efficacy, permeability, and resistance profiles remain the key open translational questions.

8. Limitations and Knowledge Gaps (of this review)

This synthesis rests on a curated set of ~29 primary and structural papers and did not independently re-derive kinetic constants or perform new sequence/structure analyses. Several claims necessarily combine data from different organisms (see §7), and the review deliberately excludes the large and heterogeneous literature on downstream rhamnosyltransferases and O-antigen polymerization, which are consumers of the module. Quantitative flux control (which step is rate-limiting in vivo, and under what growth conditions) is not resolved by the sources reviewed. The eukaryotic comparison is anchored mainly in plant (*Arabidopsis*, citrus) systems; the UDP-rhamnose routes of other eukaryotic lineages were not surveyed in depth.

9. Proposed Follow-up Experiments/Actions

1. **Species-matched kinetic panel.** Reconstitute RmlA–D from a single pathogen (e.g., *M. tuberculosis*, single-copy cassette) and measure step-by-step kinetics, cofactor preferences, and in-pathway flux control under one buffer system, removing cross-organism confounds.
2. **In vivo regulation test.** Distinguish RmlA feedback inhibition from *rml*-operon transcriptional control by combining metabolite measurements of dTDP-L-rhamnose with allele-swap experiments (feedback-resistant RmlA variants) in a live pathogen.
3. **RmlD cofactor physiology.** Determine the intracellular NADPH/NADH usage of RmlD and whether cofactor preference correlates with lineage or lifestyle.
4. **Tailoring-boundary mapping.** Systematically characterize RmlC/RmlD homologs (WbiB, EvaD, ChmJ, StrM) to define the sequence/structure determinants that switch product identity between rhamnose, 6-deoxytalose, and streptose.

5. **Whole-cell inhibitor validation.** Advance the nanomolar RmlA allosteric inhibitors and streptococcal-pathway leads from enzyme assays to whole-cell and infection models, tracking resistance emergence.

10. Key References

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